

Python for Data Science

-Final Project-

SpaceX Falcon 9 Launch Success Analysis

August 12, 2026

Table of Contents

- Executive Summary
- Introduction
- Data Collection & Wrangling
- EDA & Interactive Visualization Methodology
- EDA Results
- SQL Analysis Results
- Folium Map Results
- Plotly Dash Dashboard
- Predictive Analysis Methodology
- Predictive Analysis Results
- Conclusion
- Innovative Insights / Recommendations

Executive Summary

This project analyzes SpaceX Falcon 9 launch data to identify key factors influencing launch success rates. Data was gathered from Wikipedia using Beautiful Soup and Pandas, and analyzed using SQL.

Method

Various machine learning models—Logistic Regression, SVM, Decision Tree, and KNN—were implemented and compared to predict landing outcomes.

Findings

Launch success rates increased significantly from **2013 onward**, reaching around **80%** in recent years.

Different launch sites exhibited varying success rates, with some sites achieving higher payload capacities than others.

Certain orbit types, such as ES-L1 and GEO, had **100% success rates**, while others, like SO, showed a **0% success rate**.

Although high-payload launches were less frequent, they had a higher success proportion, indicating their reliability.

The Decision Tree model achieved the highest test accuracy of **94.4%** in predicting landing outcomes.

These insights contribute to a deeper understanding of the reliability and efficiency of reusable launches, guiding future improvements in SpaceX operations and informing strategic decisions for upcoming missions.

Introduction

- SpaceX has conducted numerous Falcon 9 launches, making the successful landing and reusing of boosters crucial for enhancing reusability and reducing launch costs. This innovative approach is essential for advancing space exploration and commercial spaceflight.
- However, the outcomes of launches and landings can vary significantly based on factors such as launch site, payload mass, and orbit type. Understanding these factors is vital for improving the reliability and efficiency of reusable launches, ultimately contributing to the sustainability of space missions.
- In this project, I analyzed historical Falcon 9 launch data to identify patterns related to launch and landing outcomes. Additionally, I built and compared various machine learning models to predict the success of booster landings, aiming to provide valuable insights for future launches.

SpaceX API (1)

Two data collection methods were used: web scraping was used to collect launch records from Wikipedia, while the SpaceX API was used to obtain detailed information linked to launch IDs.

SpaceX API

↓

requests.get() — HTTP GET Request

↓

JSON Response

↓

pd.json_normalize() — JSON → DataFrame

↓

Select relevant columns

rocket / payloads / launchpad / cores / flight_number / date_utc

↓

Data Filtering

- Remove launches other than Falacon9

<https://github.com/HaruNNH/Final-Projects/blob/main/jupyter-labs-spacex-data-collection-api.ipynb>

SpaceX API (2)



Extract information from ID-based fields

- rocket → Booster Version
- payloads → Payload Mass / Orbit
- launchpad → Launch Site / Latitude / Longitude
- cores → Landing Outcome / Flights / GridFins / Reused / Legs / Landing Pad



Data Filtering

- Remove launches other than Falacon9



Create final Pandas DataFrame



Handle missing values

- Payload Mass: replace missing values with mean

SpaceX API (3)

The following is the processed data:

	FlightNumber	Date	BoosterVersion	PayloadMass	Orbit	LaunchSite	Outcome	Flights	GridFins	Reused	Legs	LandingPad	Block	ReusedCount	Serial	Longitude	Latitude
0	1	2006-03-24	Unknown	NaN	None	None	None None	1	False	False	False	None	None	None	None	None	None
1	2	2007-03-21	Unknown	NaN	None	None	None None	1	False	False	False	None	None	None	None	None	None
2	3	2008-09-28	Unknown	NaN	None	None	None None	1	False	False	False	None	None	None	None	None	None
3	4	2009-07-13	Unknown	NaN	None	None	None None	1	False	False	False	None	None	None	None	None	None
4	5	2010-06-04	Unknown	NaN	None	None	None None	1	False	False	False	None	None	None	None	None	None

■ Data Collection (Web Scraping)

HTTP GET Request : `requests.get()` from Wikipedia



HTML Response



Beautiful Soup : `BeautifulSoup()`

This allows for easy navigation and searching within the HTML structure



Find Target Tables : `find_all("table")`

Identified the target table and get returns



Extract Table Headers & Records : `find_all("th") / find_all("tr")`



Pandas DataFrame : `pd.DataFrame()`

Date and Time Separation: Initially, date and time were combined in the same record. They were separated into two distinct columns—date and time—to facilitate analysis.

[https://github.com/HaruNNH/Final-Projects/blob/main/jupyter-labs-webscraping%20\(1\).ipynb](https://github.com/HaruNNH/Final-Projects/blob/main/jupyter-labs-webscraping%20(1).ipynb)

Data Collection (Web Scraping)

The following is the result of web scraping:

	Flight No.	Launch site	Payload	Payload mass	Orbit	Customer	Launch outcome	Version Booster	Booster landing	Date	Time
0	1	CCAFS	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	F9 v1.07B0003.1	Failure	4 June 2010	18:45
1	2	CCAFS	Dragon	0	LEO	NASA	Success	F9 v1.07B0004.1	Failure	8 December 2010	15:43
2	3	CCAFS	Dragon	525 kg	LEO	NASA	Success	F9 v1.07B0005.1	No	22 May 2012	07:44
3	4	CCAFS	SpaceX CRS-1	4,700 kg	LEO	NASA	Success	F9 v1.07B0006.1	No attempt	8 October 2012	00:35
4	5	CCAFS	SpaceX CRS-2	4,877 kg	LEO	NASA	Success	F9 v1.07B0007.1	No	1 March 2013	15:10

[https://github.com/HaruNNH/Final-Projects/blob/main/jupyter-labs-webscraping%20\(1\).ipynb](https://github.com/HaruNNH/Final-Projects/blob/main/jupyter-labs-webscraping%20(1).ipynb)

Data Wrangling Methodology

To ensure data integrity

- Null Value Check: The dataset was examined for null values to quantify their presence and impact on analysis.
- Data Type Verification: The data types were analyzed to distinguish between numerical and categorical data.

Data Transformation

- Launch Counts by Site and Orbit: An exploratory analysis was conducted to determine the number of launches and outcomes categorized by launch site and orbit type.
- Outcome Simplification: A new column named "class" was created to represent outcomes:
 - Class = 1: Successful launch
 - Class = 0: Unsuccessful launch

This transformation allowed for easier analysis and interpretation of results. The analysis revealed that the success rate across all launches was 66.7%.

Data Wrangling Methodology

The following is the result of data wrangling:

	FlightNumber	Date	BoosterVersion	PayloadMass	Orbit	LaunchSite	Outcome	Flights	GridFins	Reused	Legs	LandingPad	Block	ReusedCount	Serial	Longitude	Latitude	Class
0	1	2010-06-04	Falcon 9	6104.959412	LEO	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B0003	-80.577366	28.561857	0
1	2	2012-05-22	Falcon 9	525.000000	LEO	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B0005	-80.577366	28.561857	0
2	3	2013-03-01	Falcon 9	677.000000	ISS	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B0007	-80.577366	28.561857	0
3	4	2013-09-29	Falcon 9	500.000000	PO	VAFB SLC 4E	False Ocean	1	False	False	False	NaN	1.0	0	B1003	-120.610829	34.632093	0
4	5	2013-12-03	Falcon 9	3170.000000	GTO	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B1004	-80.577366	28.561857	0

A new column, "class," has been added to the far right, indicating whether the launch was successful or not.

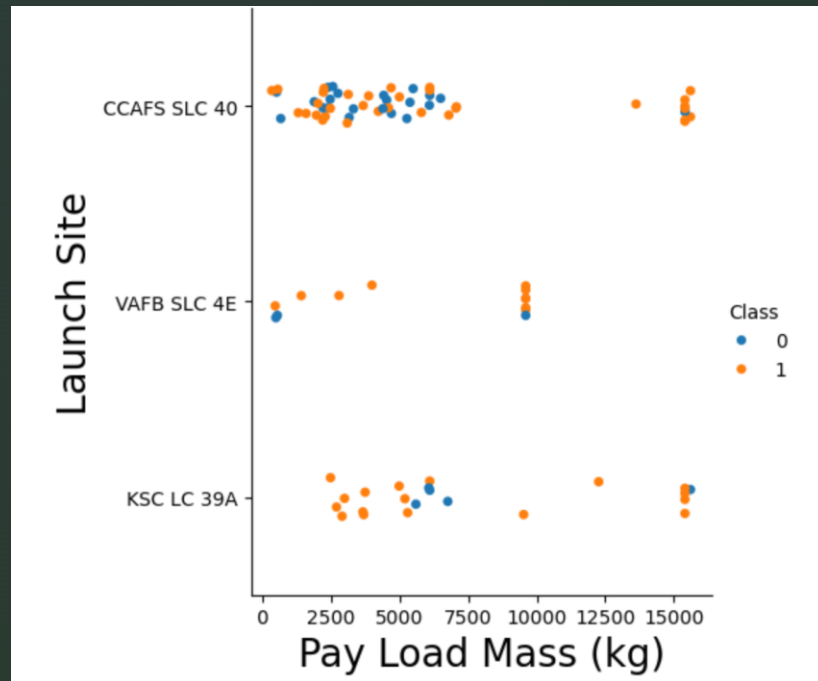
Data Cleaning

Before performing the SQL analysis, records with missing values in the Date column were removed from the dataset. These records were excluded because they did not represent valid launch records. The cleaned data was then stored in an SQLite table, SPACEXTABLE, for further analysis.

<https://github.com/HaruNNH/Final-projects/blob/main/labs-jupyter-spacex-Data%20wrangling.ipynb>

[https://github.com/HaruNNH/Final-Projects/blob/main/jupyter-labs-eda-sql-coursera_sqlite%20\(1\).ipynb](https://github.com/HaruNNH/Final-Projects/blob/main/jupyter-labs-eda-sql-coursera_sqlite%20(1).ipynb)

EDA with Data Visualization (1)



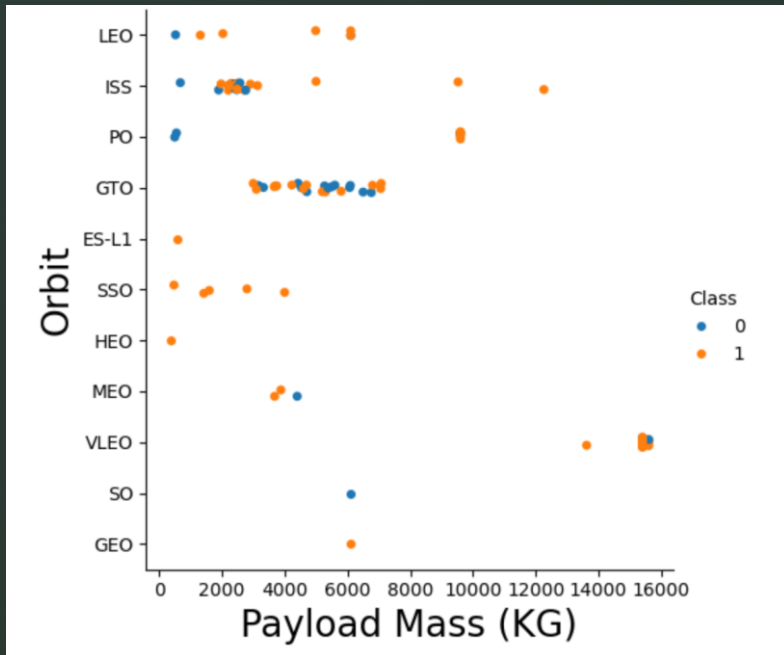
Purpose: The scatter plot illustrates the relationship between payload mass, launch site, and launch outcome (success or failure). This visualization aims to reveal how payload weight and launch site influence launch success rates.

- Smaller payloads, particularly those weighing less than 7,500 kg, were more common. However, heavier payloads appear to have a higher success rate.
- At **VAFB SLC 4E**, there were no launches with payloads exceeding 10,000 kg.
- **CCAFS SLC 40** had the highest number of launches, but it also had the highest number of failed launches among the launch sites.
- At **KSC LC 39**, launches with payloads below 2,500 kg were relatively rare.

EDA with Data Visualization (2)

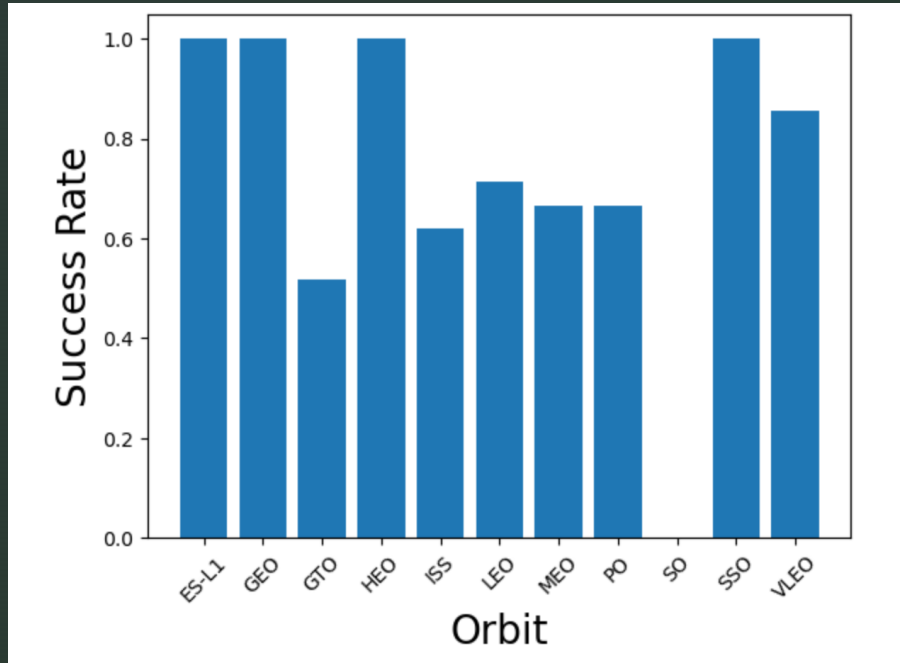
Purpose

The following scatter plot illustrates the relationship between **payload mass**, **orbit**, and **launch outcome (success or failure)**. It aims to show how payload mass and orbit selection affect launch success.



- **GEO, HEO, and ES-L1** each have only one launch attempt, making clear patterns difficult to identify.
- **GTO** has many attempts and a relatively high failure rate, with payloads mainly between **4,000 and 8,000 kg**.
- **LEO, PO, and ISS** show failures mainly with payloads below **4,000 kg**, while heavier payloads have no observed failures. **SSO** has several attempts, all successful.
- **VLEO** sometimes carries the heaviest payloads, reaching about **16,000 kg**, with successful launches more common than failures.
- Overall, **heavier payloads tend to have higher success rates, while orbit selection also affects launch outcomes.**

EDA with Data Visualization (3)



Purpose

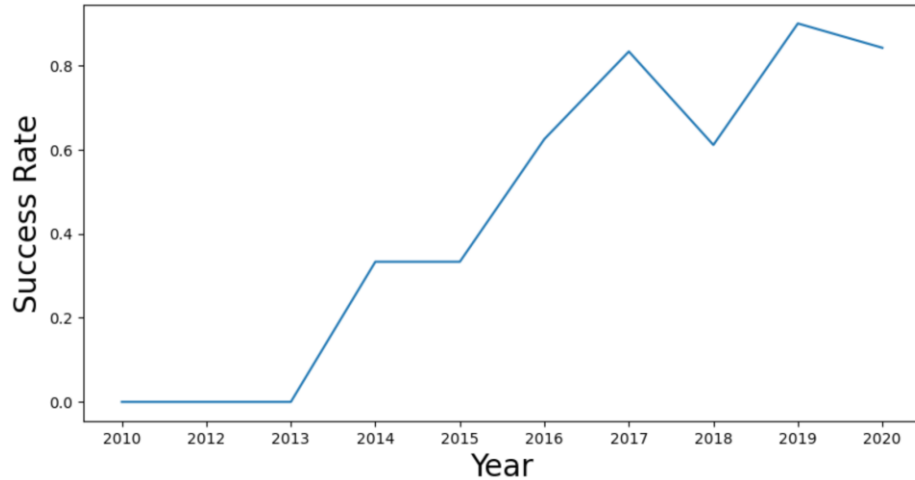
The following bar graph illustrates the relationship between orbit type and success rate. This visualization aims to identify the most appropriate orbit for launch based on the success rates recorded for each orbit, which can inform future launch strategies.

- The orbits with the highest success rates—ES-L1, GEO, HEO, and SSO—each recorded a 100% success rate, indicating their reliability for successful launches.
- Following these, VLEO achieved an 80% success rate.
- LEO has a success rate of approximately 70%, while MEO and PO have similar rates of slightly over 60%. ISS recorded a 60% success rate, and GTO had a success rate of around 50%.
- Notably, SO had the lowest success rate at 0%, highlighting potential challenges associated with this orbit.

EDA with Data Visualization (3)

Purpose

The graph shows changes in launch success rates over time, highlighting the improvement in SpaceX's launch reliability. This analysis helps identify trends in performance and factors related to launch success.



The success rate generally increased from 2013 to 2019, with a significant rise of about 30 percentage points from 2013 to 2014. The rate remained stable in 2015, then increased to around 60% in 2016. Notably, it first reached **80% in 2017**, but experienced a drop to 60% in 2018 before reaching its highest success rate of **over 80% in 2019**.

This fluctuation highlights the challenges faced in 2018, which may warrant further investigation into factors that affected launch outcomes that year. Overall, the trend indicates a positive trajectory in SpaceX's launch success rates over time.

SQL Analysis Results

Purpose: Display the names of the unique launch sites in the space mission

SQL example:

```
pd.read_sql("SELECT DISTINCT Launch_Site FROM SPACEXTABLE", con)
```

	Launch_Site
0	CCAFS LC-40
1	VAFB SLC-4E
2	KSC LC-39A
3	CCAFS SLC-40

Purpose: List the total number of successful and failure mission outcomes

SQL example:

```
pd.read_sql("""
SELECT "Mission_Outcome", COUNT(*) AS count
FROM SPACEXTABLE
GROUP BY "Mission_Outcome"
""", con)
```

	Mission_Outcome	count
0	Failure (in flight)	1
1	Success	98
2	Success	1
3	Success (payload status unclear)	1

SQL Analysis Results

Purpose: Display the total payload mass carried by boosters launched by NASA (CRS)

SQL example:

```
pd.read_sql("SELECT SUM(PAYLOAD_MASS__KG_) AS total_payload_mass FROM SPACE_TABLE WHERE Customer = 'NASA (CRS)'", con)
```

total_payload_mass	
0	45596

Purpose: Display average payload mass carried by booster version F9 v1.1

SQL example:

```
pd.read_sql("SELECT AVG(PAYLOAD_MASS__KG_) AS average_payload_mass FROM SPACE_TABLE WHERE version = 'F9 v1.1'", con)
```

average_payload_mass	
0	2928.4

SQL Analysis Results

Purpose: List the date when the first succesful landing outcome in ground pad was acheived.

SQL example:

```
pd.read_sql("SELECT MIN(Date) AS first_successful_ground_landing FROM SPACEXTABLE  
WHERE Landing_Outcome = 'Success (ground pad)'", con)
```

first_successful_ground_landing	
0	2015-12-22

SQL Analysis Results

Purpose: Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order

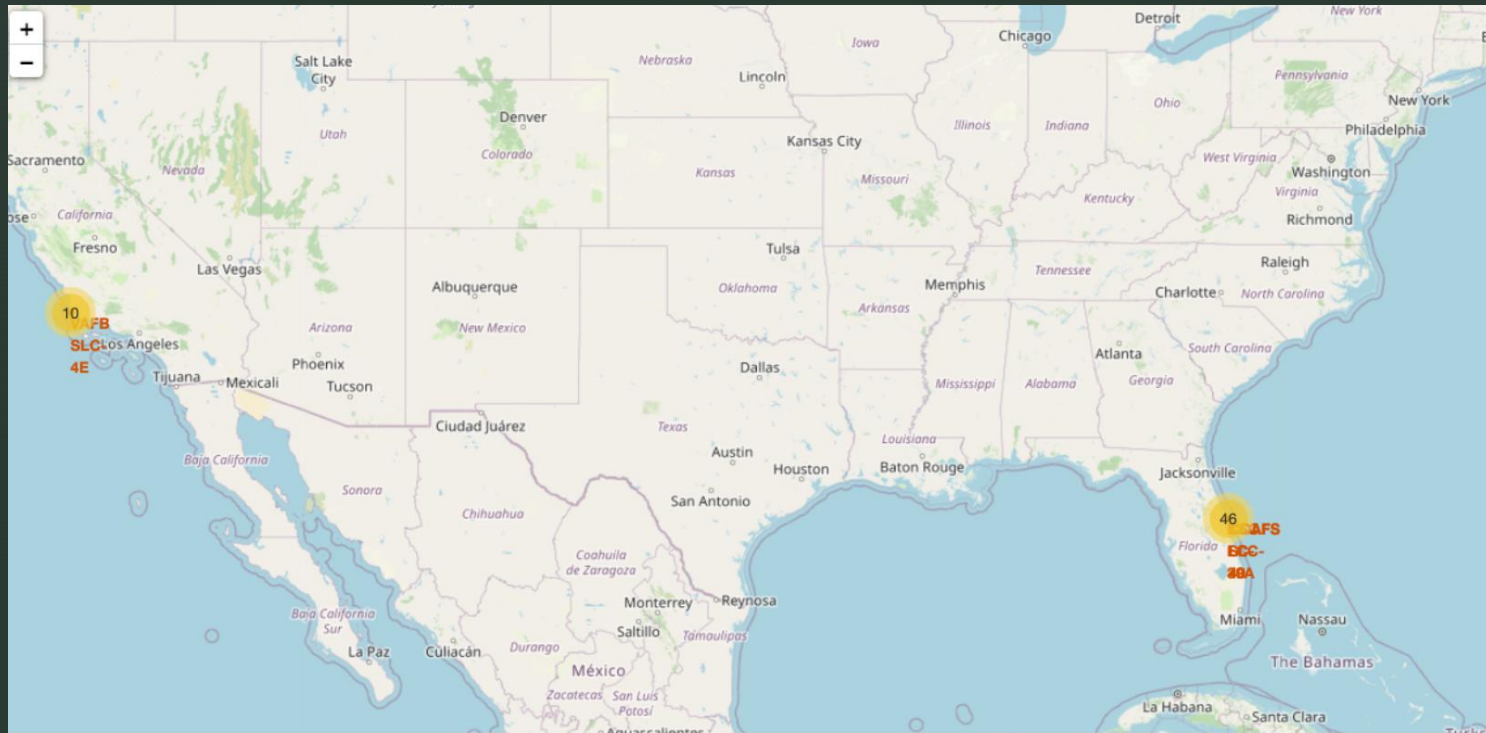
SQL example:

```
pd.read_sql("""
SELECT "Landing_Outcome", COUNT(*) AS count
FROM SPACEXTABLE
WHERE Date BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY "Landing_Outcome"
ORDER BY count DESC
""", con)
```

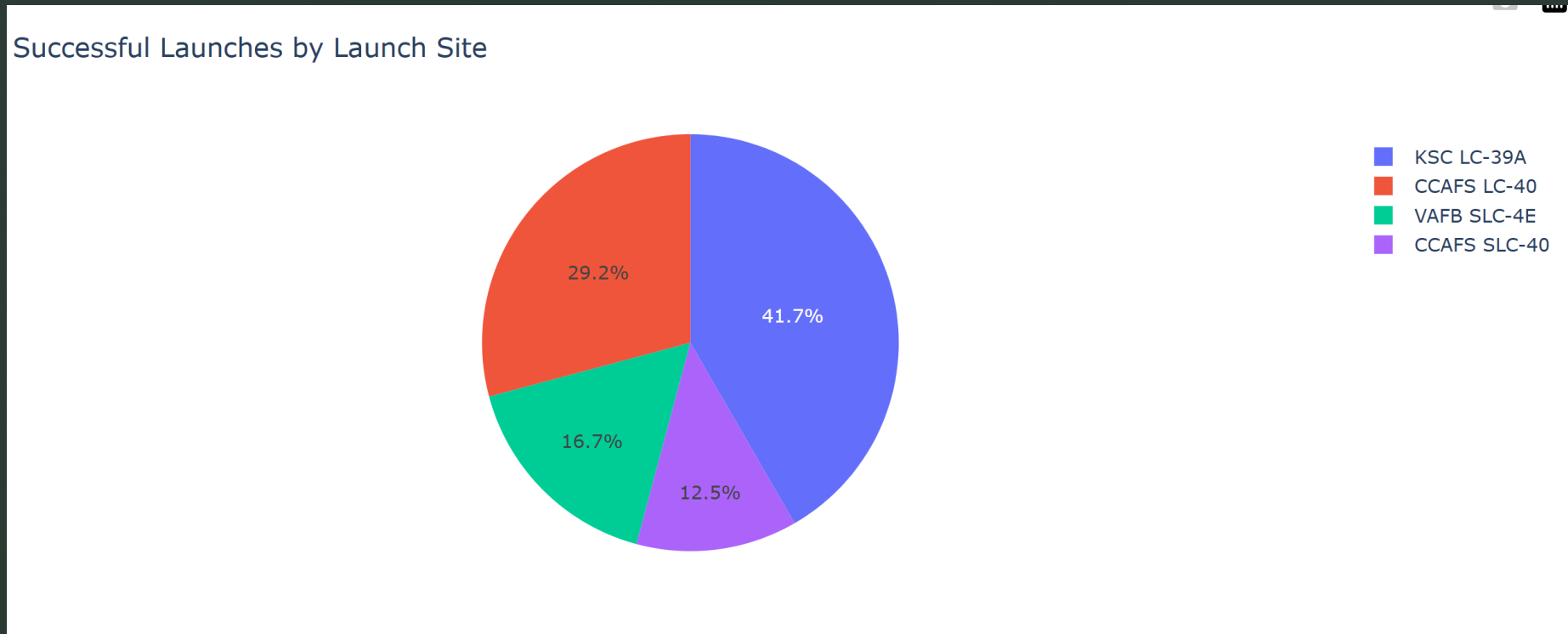
	Landing_Outcome	count
0	No attempt	10
1	Success (drone ship)	5
2	Failure (drone ship)	5
3	Success (ground pad)	3
4	Controlled (ocean)	3
5	Uncontrolled (ocean)	2
6	Failure (parachute)	2
7	Precluded (drone ship)	1

Folium Map Results

The dataset contains 56 launch records from three launch sites. Each record includes the launch site, geographic coordinates, and launch success class. The map shows that most launches took place at sites on the east coast.



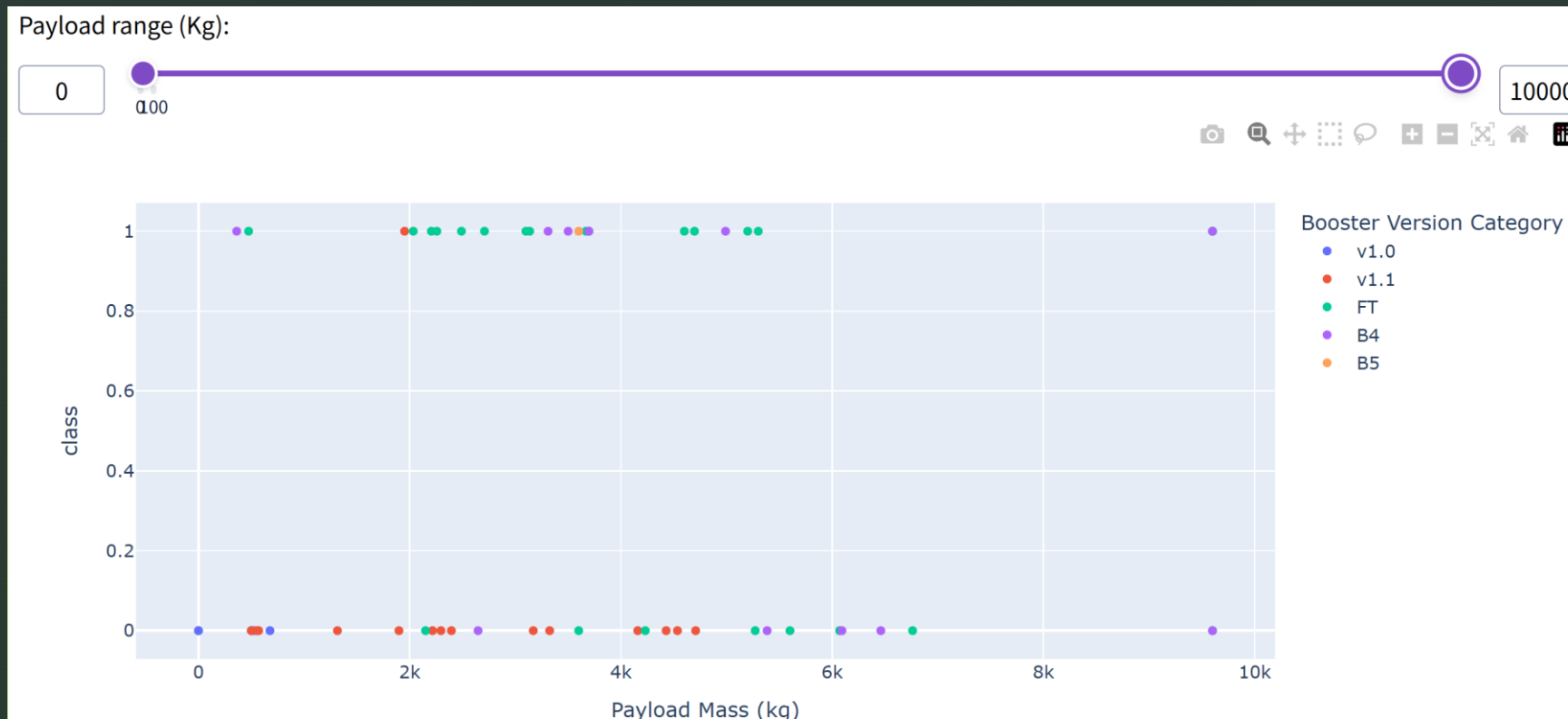
Plotly Dash Dashboard (1)



**This chart shows the proportion of successful launches by launch site.
KSC LC-39A has the highest number of successful launches.**

<http://github.com/HaruNNH/Final-Projects/blob/main/spacex-dash-app.py>

Plotly Dash Dashboard (2)



By booster version, FT and B4 have the highest number of successful launches, while failures are more common with V1.1. Launches carrying more than 8,000 kg used the B4 booster.

<http://github.com/HaruNNH/Final-Projects/blob/main/spacex-dash-app.py>

Predictive Analysis Methodology

1. Data Preprocessing

Class was used as the target variable.

Features were standardized using StandardScaler.

The data was split into training and test sets (80% / 20%).

2. Model Training

Logistic Regression

Support Vector Machine (SVM)

Decision Tree

K-Nearest Neighbors (KNN)

3. Hyperparameter Tuning

GridSearchCV with 10-fold cross-validation was used to select the optimal hyperparameterset.

4. Model Evaluation

Test accuracy

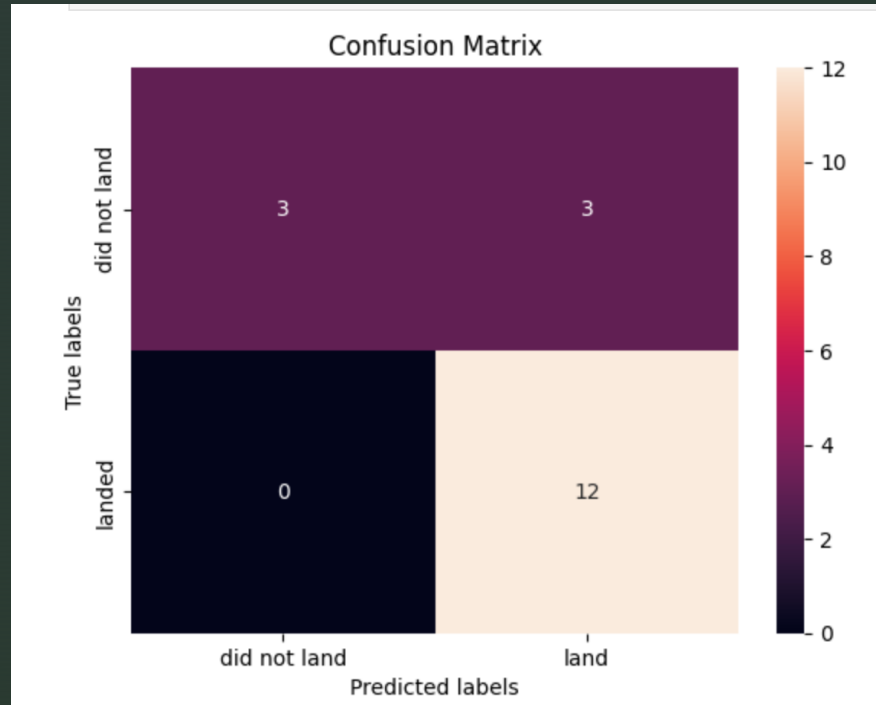
Confusion matrix

5. Model Comparison

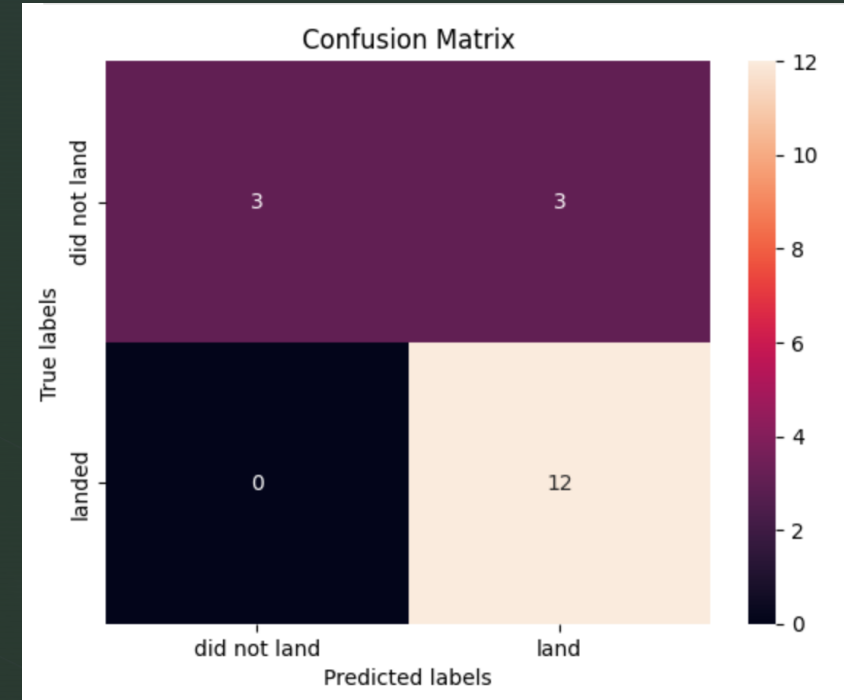
The performance of the four classification models was compared using test-set accuracy.

Predictive Analysis Results

Logistic Regression

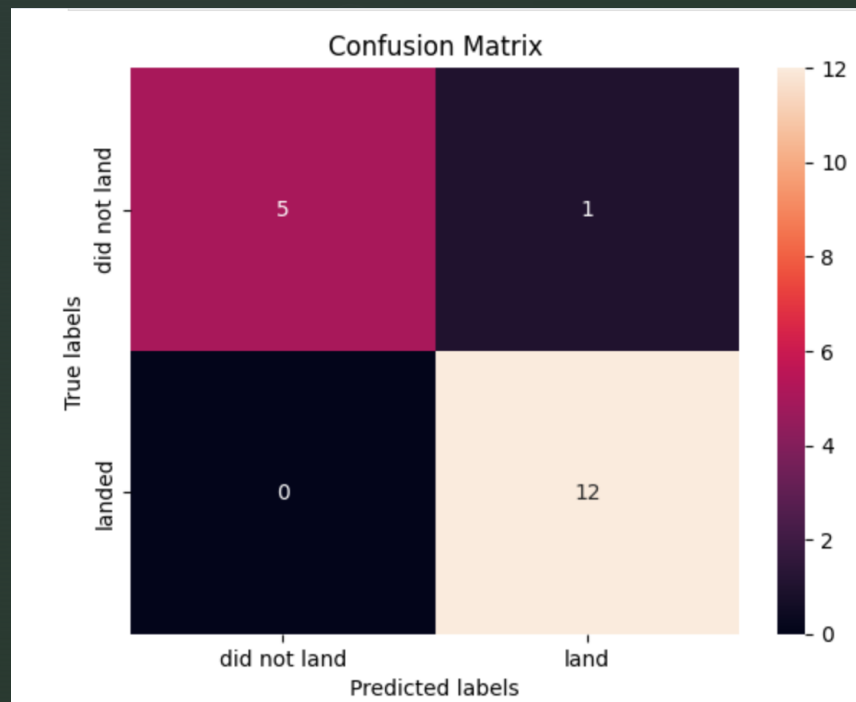


SVM

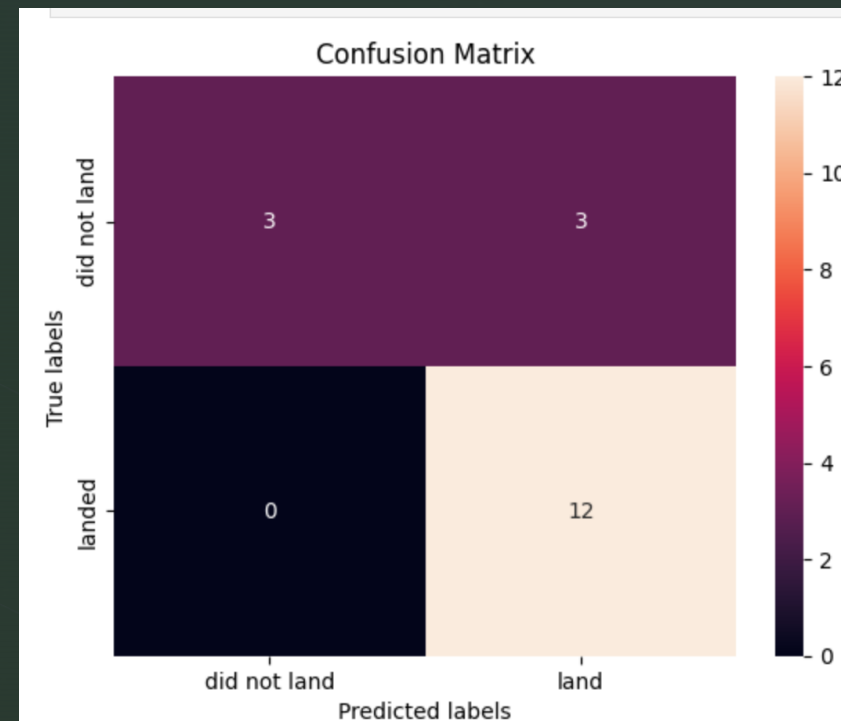


Predictive Analysis Results

Tree Decision



KNN



Predictive Analysis Results

The Decision Tree was the most accurate model among the four models, correctly predicting 17 out of 18 test cases and achieving an accuracy of 94.4%.

The following table shows the test accuracies

Model	Accuracy
Logistic Regression	83.3%
SVM	83.3%
Decision Tree	94.4%
KNN	83.3%

Overall, the Decision Tree performed best on the test data.

Conclusion

The analysis showed that Falcon 9 launch success is related to several factors, including launch site, orbit, payload mass, and booster version.

The exploratory and geographical analyses revealed differences in launch success across sites and booster versions.

The machine learning analysis showed that the Decision Tree performed best among the four models, achieving 94.4% accuracy on the test data.

Overall, this analysis suggests that launch characteristics and booster technology are important factors in predicting Falcon 9 landing success. However, the dataset is relatively small, so the results should be interpreted with caution.

■ Innovative Insights / Recommendations

Since booster reusability is important for reducing launch costs, improving the landing success rate is essential for SpaceX.

Collecting more launch data in the future could improve the accuracy of the analysis and predictions. This may help identify the factors that affect landing success more clearly and ultimately contribute to higher landing and overall mission success rates.

Access to my GitHub

<https://github.com/HaruNNH/Final-Projects>